

BCNF Analysis

1 BCNF Check

Consider the relations in the ERD diagram for GPU Inference Routing.

Region

$region_id \rightarrow \{region_name, country\}$

InferenceRequest

$request_id \rightarrow \{submitted_at, input_size, status, region_id, machine_id, model_id, response_time_ms\}$

AIModel

$model_id \rightarrow \{model_name\}$

Hosts

$(machine_id, model_id) \rightarrow$

GPUServer

$machine_id \rightarrow \{cpu_capacity, memory_capacity, gpu_capacity, gpu_type_name\}$

MachineMetric

$metric_id \rightarrow \{start_time, end_time, worker_load, cpu_utilization, gpu_utilization\}$

GPUType

$gpu_type_name \rightarrow \{memory_gb\}$

CompatibleWith

$(model_id, gpu_type_name) \rightarrow$

All determinants are superkeys, so all relations satisfy BCNF.